

investors having superior due diligence capabilities. Even these investors may want to reconsider their level of commitment to these alternative investments. Barber and Wang (2011) did a 20-year study of university endowment performance from 1991 through 2011. They found that for the average endowment, factor models with only stock and bond benchmarks explain virtually all of the time-series variation in their returns and show no alpha.

ACTIVELY MANAGED MUTUAL FUNDS

We have seen that it is difficult for investors to profit from funds that charge annual fees of 2% of assets and 20% of profits, such as managed futures, hedge funds, and private equity. Let us look now at other forms of active management that do not charge such high fees.

More than 52 million households own mutual funds. Total U.S. assets in mutual funds now exceed \$11.6 trillion. According to the Investment Company Institute (ICI), index funds make up 17.4% of domestic equity mutual funds, while actively managed equity funds make up 82.6%.

Researchers have extensively studied the performance of actively managed mutual funds since the 1960s. Jensen (1968) helped passive management gain a foothold with his study showing that the average mutual fund from 1945 through 1964 did no better than buying and holding the market.

Survivorship bias was a significant issue in these early studies. Through 2012, only 51% of actively managed mutual funds had survived over the preceding 10 years. One of the first comprehensive studies of mutual fund performance that considered survivorship bias was by Malkiel (1995). He found that, in aggregate, equity mutual funds from 1971 through 1991 underperformed their benchmark portfolios, both after and before management expenses.

Fama and French (2009) picked up where Malkiel left off. Using mutual fund data from 1984 through 2006, they discovered that few actively managed funds produced benchmark-adjusted returns sufficient to cover their costs, and it was hard to tell whether good performance was due to skill or to luck. The only thing predictable was that actively managed funds with